EE922: 5G Wireless Standards Design **EE Dept. IITK**

MATLAB Assignment-3

5G Transceiver Implementation -II

The objective of this assignment is to implement rate matching and rate recovery modules. These modules should be integrated with the existing transceiver modules. Your code should contain

1. Transmit chain which will
   - Generate a transport block (TB) of size 20496 and append TB-CRC
   - Segment the TB and calculate and append the CB-CRC (implement the segmentation from the standard)
   - LDPC encoding for all the segmented code blocks (Interleaving and interlacing are not included)
   - Perform rate matching for MCS 9 by considering an allocation of 100 PRBs over a slot of 14 symbols. You can assume that six REs in a slot are reserved for pilots. Assume that RV0 is being transmitted.
   - Perform code block concatenation
   - Perfom QPSK modulation according to the standard. You should remove the BPSK modulator (mod_output) provided with the LDPC encoder
2. Receive chain which will
   - Perfom QPSK demodulation (create a new QPSK demodulator function by modifying the BPSK demodulator provided with LDPC decoder)
   - Perform code block segmentation
   - Perform rate recovery for MCS 9 assuming RV0
   - Perform LDPC decoding for each segmented code block
   - Validate and remove the CB-CRC for each code block
   - Concatenate the segemented decoder code blocks
   - Show that the transmit and receive code block match

You have to only implement the modules marked in red.
Please follow these Coding instructions:

- Properly comment your code.
- The code should execute and generate the desired output.
- Your submission should be self-contained (should include all the files required for running it).
- Avoid hard-coding the values of the variables for specific configurations. The code should be generic.

Please follow these submission instructions.

- Deadline is 1st of June, 11:59 pm.
- All codes should be in one .zip/.rar folder. Please do not submit separate files.
- Upload your properly commented in drive link which is provided to you. Name your code as rollno.zip.
- Please submit one final zip file.
- Please do not mail your file to me.

Please also read this carefully.

- Each one of you have to individually do all the reading and MATLAB assignments. You can discuss with your friends but you will have to completely write your own code.

Multiplexing and Channel Coding

Detailed explanation of the **5G NR transceiver project** aligned with the 3GPP TS38.212-f20 standard, broken into **logical functional blocks** with clear explanations of **what each stage does and why**.

🎯 Project Objective

To simulate a complete **5G NR Transmitter and Receiver chain**, including:

- CRC generation
- Code block segmentation
- LDPC channel coding
- Rate matching
- Modulation and demodulation
- Receiver-side decoding and CRC validation

🧱 Project Structure Overview

1. User Inputs

- **Transport Block (TB)**: Can be manually entered or randomly generated.
- **CRC Polynomial**: Chosen by user from allowed options (e.g., CRC24A/B/C, CRC16).
- **Modulation and Coding Scheme (MCS)**: Selects modulation order and code rate.

🔐 2. CRC Addition

Purpose: Error detection at TB and Code Block (CB) levels.

- **Transport Block CRC** is calculated and appended using the selected generator polynomial.
- **CB CRC** is added after segmentation (for each segment) using another polynomial.

✂️ 3. Code Block Segmentation

Why: TBs can be too large for LDPC encoders; hence they are segmented.

- TB is split into multiple **code blocks (CBs)** based on maximum allowable size (**K_cb**) depending on chosen **Base Graph (BG1 or BG2)**.
- Filler bits are added to match the exact size needed by the LDPC matrix.

🧬 4. LDPC Encoding

Why: LDPC (Low-Density Parity-Check) codes are used for error correction.

- Based on **BG1 (for large blocks, higher rates)** or **BG2 (for small blocks, lower rates)**.
- Uses MATLAB's LDPCEncode() for each CB.

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5. Rate Matching

Why: Match the LDPC codeword length to available resources in the physical layer.

- Achieved by:
    - **Puncturing** (removing bits),
    - **Repetition** (duplicating bits),
    - **Interleaving**.
- Controlled via **Redundancy Versions (RV0–RV3)** and circular buffering.

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6. Modulation

Why: Convert bits to symbols for physical transmission.

- **Modulation schemes**: BPSK and QPSK supported.
- **Resource allocation**: Number of PRBs × REs × Mod Order = total transmittable bits (**G**).

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7. AWGN Channel Simulation

- Adds optional **Gaussian noise** to the transmitted symbols.

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8. Receiver Side

8.1 Demodulation

- Converts noisy symbols back into **LLRs** (Log-Likelihood Ratios).

8.2 Rate Recovery

- Inverse of rate matching: Reconstructs full-length LDPC codewords from rate-matched data.

8.3 LDPC Decoding

- Decodes each CB using LDPCDecode().

8.4 CRC Validation

- Validates CB CRC and TB CRC.
- If CRC fails: program exits with error message.

Notable Features

- Supports **Base Graph switching**.
- Interactive **user prompts** for input and MCS settings.
- Extensive **debug output** showing lengths, offsets, and errors.
- Practical for **bit-level simulation** of PHY layer.

📚 Standards Referenced

- **3GPP TS 38.212** – Channel coding (LDPC, CRC, segmentation).
- **3GPP TS 38.214** – MCS tables and spectral efficiencies.

💡 How You Can Use or Extend This

- Add support for **higher-order modulations** (e.g., 16QAM, 64QAM).
- Include **HARQ combining** logic using different RVs.
- Visualize BER vs SNR curves by adding Eb/No loop.
- Integrate this with MATLAB 5G Toolbox or Simulink models.

✅ Summary

This project is a **hands-on implementation** of a complete **5G NR physical layer chain**. It:

- Validates the **3GPP-compliant LDPC coding and rate matching** process.
- Shows how CRC, segmentation, and modulation interact.
- Is suitable for both **learning and interview preparation** for PHY/MAC roles.
